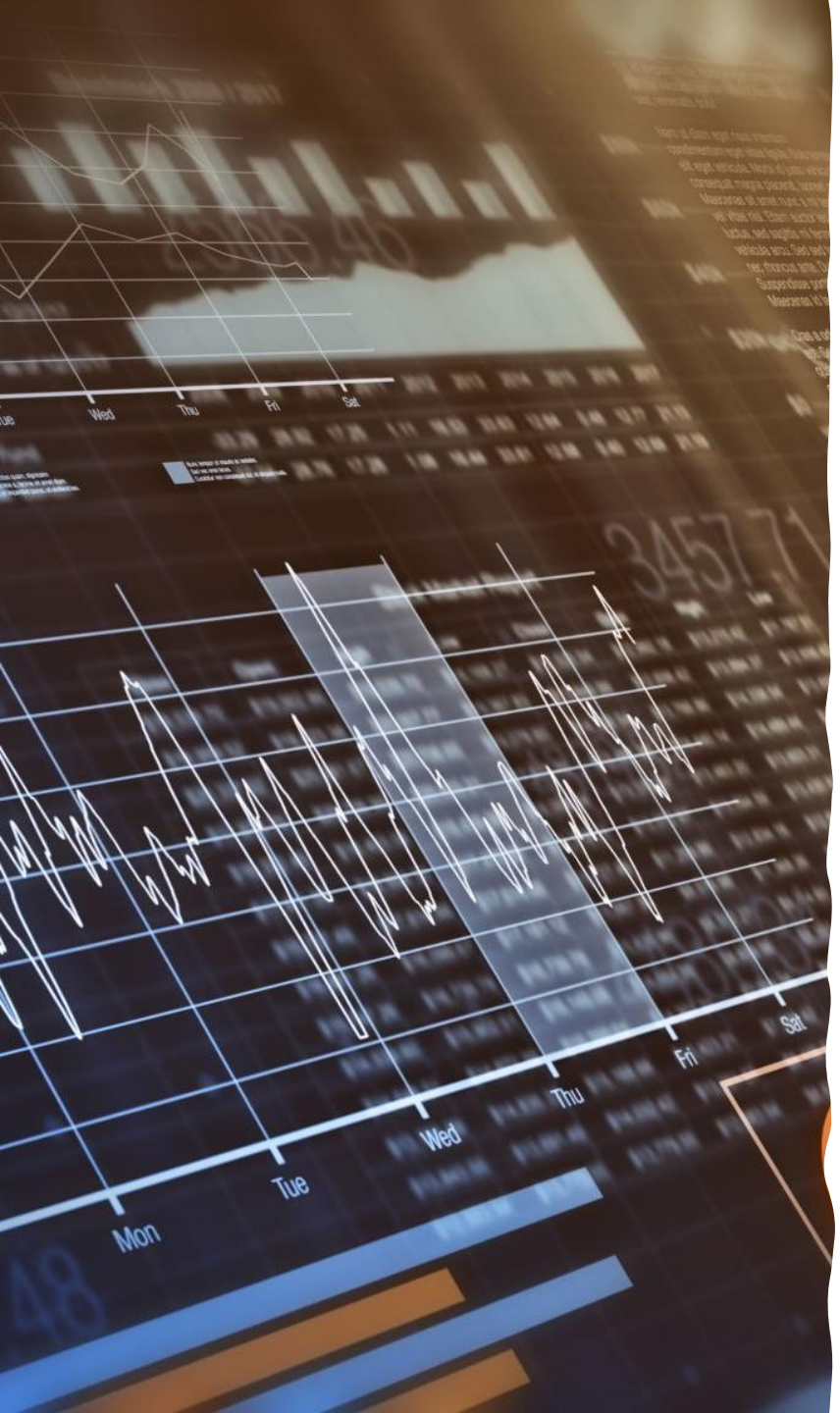
The Technology Value Stream

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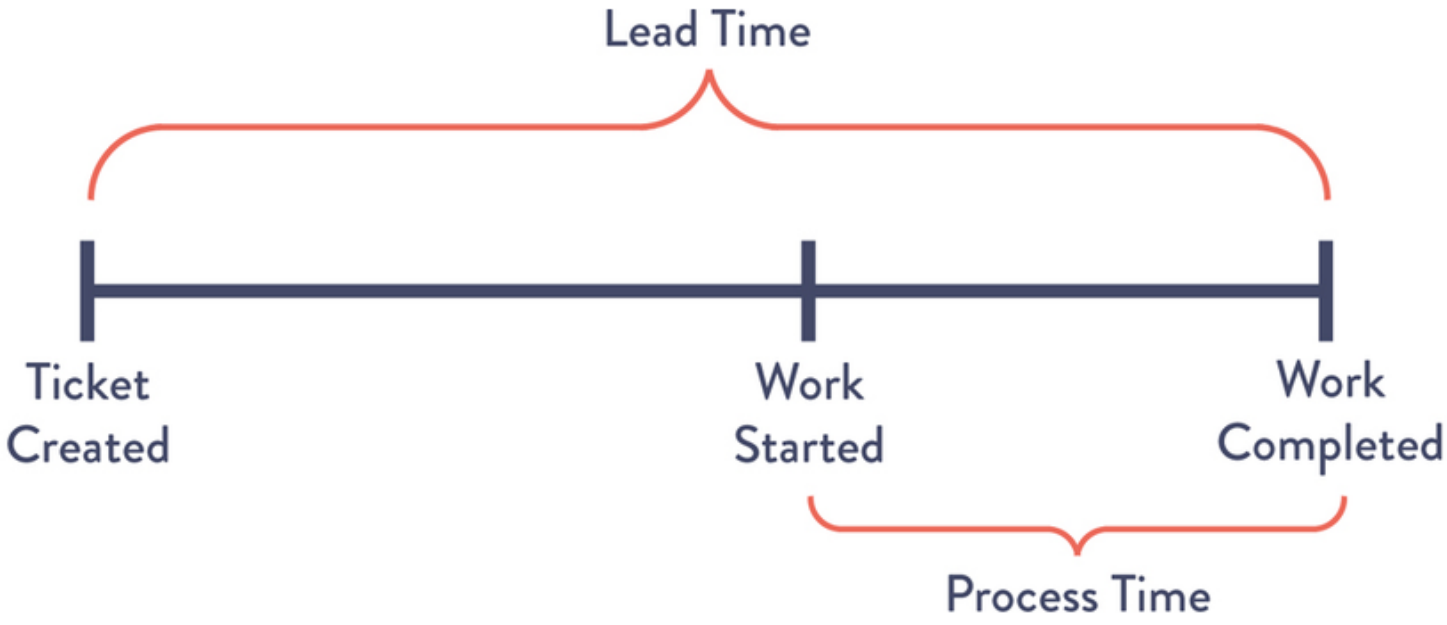


What is the Technology Value Stream?

- “the process required to convert a business hypothesis into a technology-enabled service or feature that delivers value to the customer” (Kim et al., 2016)
- The Value Stream starts as soon as a customer makes a request and it is added to the future plans

Lead Time vs Process Time

- **Lead time:** Total time elapsed from a request being made until it is completed
- **Process time:** Time elapsed from when work starts until it is completed



The Common Scenario: Deployment Takes Months

- Teams work separately and pass work between departments (Waterfall development)
- Testing, approvals, and deployment may be manual
- Merging code can lead to lengthy investigations if there are errors
- Releases become large, infrequent, and risky

The DevOps Approach on Value Stream

- More efficient development
- improve response & deployment time
- better customer experience





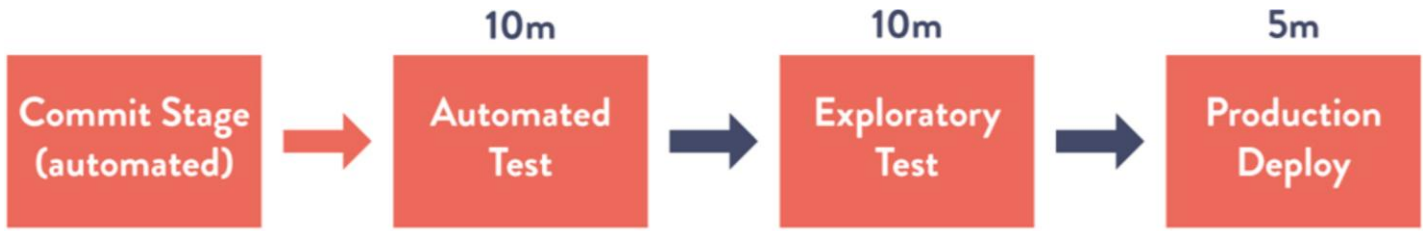
The DevOps Ideal: Deployment in Minutes

AUTOMATE BUILDING, TESTING, AND DEPLOYMENT

INTEGRATE DEVELOPMENT OPERATIONS

MAKE SMALLER, MORE FREQUENT CHANGES

IDENTIFY PROBLEMS QUICKLY AND RECEIVE FEEDBACK SOONER

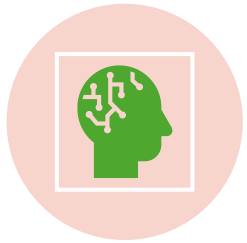


Automatic Approval

Manual Approval

Total Cycle Time: 25m

Reducing Lead Time



AUTOMATE REPETITIVE TASKS



USE CONTINUOUS INTEGRATION AND CONTINUOUS DELIVERY (CI / CD)



REDUCE UNNECESSARY HANDOFFS AND BOTTLENECKS



SHARE RESPONSIBILITY BETWEEN DEVELOPMENT AND OPERATIONS



USE MONITORING AND FEEDBACK TO IMPROVE THE PROCESS

Why this matters

- Earlier delivery of features and updates
- Earlier feedback leading to earlier fixes
- Smaller changes are easier to troubleshoot
- Faster recovery through appropriate testing and monitoring





References

Kim, G., Debois, P., Willis, J., & Humble, J. (2016). The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations. *Agile, Continuous Delivery, and the Three Ways*.

Quiroz-Vázquez, C. (n.d.). *What is value stream management?* Retrieved October 25, 2026, from https://www.ibm.com/think/topics/value-stream-management

Takakura, T. (n.d.). *What is Value Stream Management (VSM) and Why is it Important?* Atlassian. Retrieved October 25, 2026, from https://www.atlassian.com/agile/value-stream-management